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- Abramowitz, M., and Stegun, I.A. 1964, *Handbook of Mathematical Functions* (Washington: National Bureau of Standards); reprinted 1968 (New York: Dover); online at <http://www.nist.gov/aands>, Chapter 5.
- Wrench J.W. 1952, “A New Calculation of Euler’s Constant,” *Mathematical Tables and Other Aids to Computation*, vol. 6, p. 255.[2]

6.4 Incomplete Beta Function

The incomplete beta function is defined by

$$I_x(a, b) \equiv \frac{B_x(a, b)}{B(a, b)} \equiv \frac{1}{B(a, b)} \int_0^x t^{a-1} (1-t)^{b-1} dt \quad (a, b > 0) \quad (6.4.1)$$

It has the limiting values

$$I_0(a, b) = 0 \quad I_1(a, b) = 1 \quad (6.4.2)$$

and the symmetry relation

$$I_x(a, b) = 1 - I_{1-x}(b, a) \quad (6.4.3)$$

If a and b are both rather greater than one, then $I_x(a, b)$ rises from “near-zero” to “near-unity” quite sharply at about $x = a/(a+b)$. Figure 6.4.1 plots the function for several pairs (a, b) .

The incomplete beta function has a series expansion

$$I_x(a, b) = \frac{x^a (1-x)^b}{a B(a, b)} \left[1 + \sum_{n=0}^{\infty} \frac{B(a+1, n+1)}{B(a+b, n+1)} x^{n+1} \right] \quad (6.4.4)$$

but this does not prove to be very useful in its numerical evaluation. (Note, however, that the beta functions in the coefficients can be evaluated for each value of n with just the previous value and a few multiplies, using equations 6.1.9 and 6.1.3.)

The continued fraction representation proves to be much more useful:

$$I_x(a, b) = \frac{x^a (1-x)^b}{a B(a, b)} \left[\frac{1}{1 + \frac{d_1}{1 + \frac{d_2}{\dots}}} \right] \quad (6.4.5)$$

where

$$\begin{aligned} d_{2m+1} &= -\frac{(a+m)(a+b+m)x}{(a+2m)(a+2m+1)} \\ d_{2m} &= \frac{m(b-m)x}{(a+2m-1)(a+2m)} \end{aligned} \quad (6.4.6)$$

This continued fraction converges rapidly for $x < (a+1)/(a+b+2)$, except when a and b are both large, when it can take $O(\sqrt{\min(a, b)})$ iterations. For $x >$

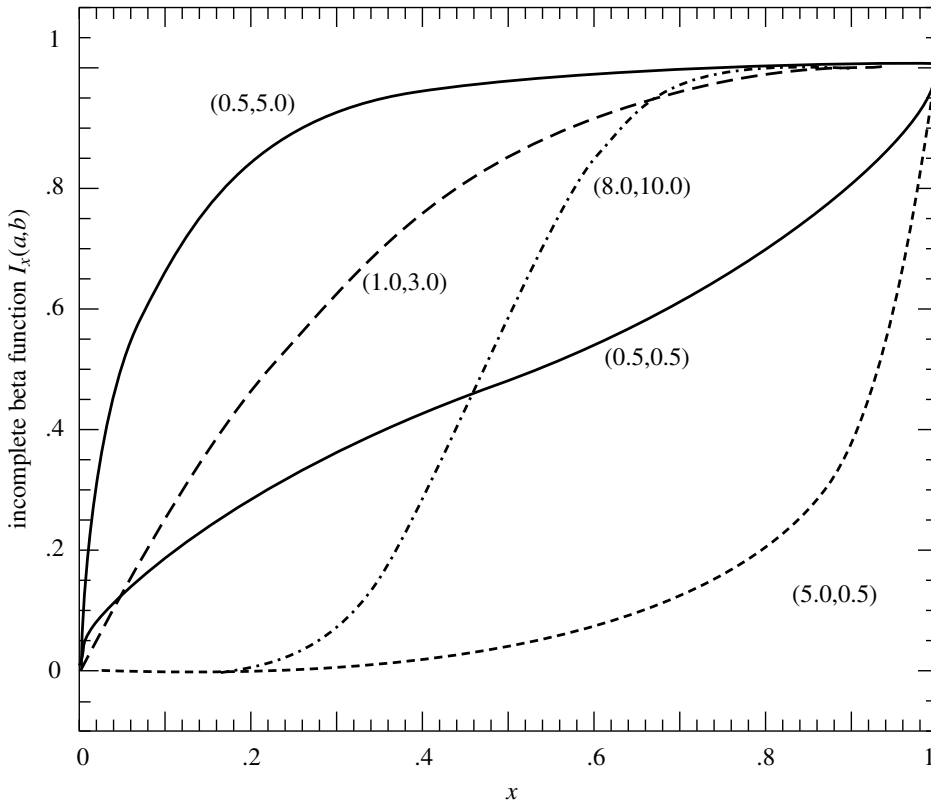


Figure 6.4.1. The incomplete beta function $I_x(a, b)$ for five different pairs of (a, b) . Notice that the pairs $(0.5, 5.0)$ and $(5.0, 0.5)$ are symmetrically related as indicated in equation (6.4.3).

$(a + 1)/(a + b + 2)$ we can just use the symmetry relation (6.4.3) to obtain an equivalent computation in which the convergence is again rapid. Our computational strategy is thus very similar to that used in Gamma: We use the continued fraction except when a and b are both large, in which case we do a single step of high-order Gauss-Legendre quadrature.

Also as in Gamma, we code an inverse function using Halley's method. When a and b are both ≥ 1 , the initial guess comes from §26.5.22 in reference [1]. When either is less than 1, the guess comes from first crudely approximating

$$\int_0^1 t^{a-1}(1-t)^{b-1} dt \approx \frac{1}{a} \left(\frac{a}{a+b} \right)^a + \frac{1}{b} \left(\frac{b}{a+b} \right)^b \equiv S \quad (6.4.7)$$

which comes from breaking the integral at $t = a/(a+b)$ and ignoring one factor in the integrand on each side of the break. We then write

$$I_x(a, b) \approx \begin{cases} x^a/(Sa) & x \leq a/(a+b) \\ (1-x)^b/(Sb) & x > a/(a+b) \end{cases} \quad (6.4.8)$$

and solve for x in the respective regimes. While crude, this is good enough to get well within the basin of convergence in all cases.

ncgammabeta.h

struct Beta : Gauleg18 {

Object for incomplete beta function. Gauleg18 provides coefficients for Gauss-Legendre quadrature.

```
static const Int SWITCH=3000;           When to switch to quadrature method.
static const Doub EPS, FPMIN;           See end of struct for initializations.
```

```
Doub betai(const Doub a, const Doub b, const Doub x) {
```

Returns incomplete beta function $I_x(a, b)$ for positive a and b , and x between 0 and 1.

```
    Doub bt;
    if (a <= 0.0 || b <= 0.0) throw("Bad a or b in routine betai");
    if (x < 0.0 || x > 1.0) throw("Bad x in routine betai");
    if (x == 0.0 || x == 1.0) return x;
    if (a > SWITCH && b > SWITCH) return betaiapprox(a,b,x);
    bt=exp(gammln(a+b)-gammln(a)-gammln(b)+a*log(x)+b*log(1.0-x));
    if (x < (a+1.0)/(a+b+2.0)) return bt*betacf(a,b,x)/a;
    else return 1.0-bt*betacf(b,a,1.0-x)/b;
}
```

```
Doub betacf(const Doub a, const Doub b, const Doub x) {
```

Evaluates continued fraction for incomplete beta function by modified Lentz's method (§5.2). User should not call directly.

```
    Int m,m2;
    Doub aa,c,d,del,h,qab,qam,qap;
    qab=a+b;
    qap=a+1.0;
    qam=a-1.0;
    c=1.0;
    d=1.0-qab*x/qap;
    if (fabs(d) < FPMIN) d=FPMIN;
    d=1.0/d;
    h=d;
    for (m=1;m<10000;m++) {
        m2=2*m;
        aa=m*(b-m)*x/((qam+m2)*(a+m2));
        d=1.0+aa*d;
        if (fabs(d) < FPMIN) d=FPMIN;
        c=1.0+aa/c;
        if (fabs(c) < FPMIN) c=FPMIN;
        d=1.0/d;
        h *= d*c;
        aa = -(a+m)*(qab+m)*x/((a+m2)*(qap+m2));
        d=1.0+aa*d;
        if (fabs(d) < FPMIN) d=FPMIN;
        c=1.0+aa/c;
        if (fabs(c) < FPMIN) c=FPMIN;
        d=1.0/d;
        del=d*c;
        h *= del;
        if (fabs(del-1.0) <= EPS) break;
    }
    return h;
}
```

These q's will be used in factors that occur in the coefficients (6.4.6).

First step of Lentz's method.

One step (the even one) of the recurrence.

Next step of the recurrence (the odd one).

Are we done?

```
Doub betaiapprox(Doub a, Doub b, Doub x) {
```

Incomplete beta by quadrature. Returns $I_x(a, b)$. User should not call directly.

```
    Int j;
    Doub xu,t,sum,ans;
    Doub a1 = a-1.0, b1 = b-1.0, mu = a/(a+b);
    Doub lnmu=log(mu),lnmuc=log(1.-mu);
    t = sqrt(a*b/(SQR(a+b)*(a+b+1.0)));
    if (x > a/(a+b)) {
        if (x >= 1.0) return 1.0;
        xu = MIN(1.,MAX(mu + 10.*t, x + 5.0*t));
    } else {
```

Set how far to integrate into the tail:

```

    if (x <= 0.0) return 0.0;
    xu = MAX(0., MIN(mu - 10.*t, x - 5.0*t));
}
sum = 0;
for (j=0; j<18; j++) {
    t = x + (xu-x)*y[j];
    sum += w[j]*exp(a1*(log(t)-lnmu)+b1*(log(1-t)-lnmuc));
}
ans = sum*(xu-x)*exp(a1*lnmu-gammln(a)+b1*lnmuc-gammln(b)+gammln(a+b));
return ans>0.0? 1.0-ans : -ans;
}

Doub invbetai(Doub p, Doub a, Doub b) {
    Inverse of incomplete beta function. Returns  $x$  such that  $I_x(a, b) = p$  for argument  $p$ 
    between 0 and 1.
    const Doub EPS = 1.e-8;
    Doub pp, t, u, err, x, al, h, w, afac, a1=a-1., b1=b-1.;
    Int j;
    if (p <= 0.) return 0.;
    else if (p >= 1.) return 1.;
    else if (a >= 1. && b >= 1.) {
        Set initial guess. See text.
        pp = (p < 0.5)? p : 1. - p;
        t = sqrt(-2.*log(pp));
        x = (2.30753+t*0.27061)/(1.+t*(0.99229+t*0.04481)) - t;
        if (p < 0.5) x = -x;
        al = (SQR(x)-3.)/6.;
        h = 2./(1./(2.*a-1.))+1./(2.*b-1.);
        w = (x*sqrt(al+h)/h)-(1./(2.*b-1)-1./(2.*a-1.))*(al+5./6.-2./(3.*h));
        x = a/(a+b*exp(2.*w));
    } else {
        Doub lna = log(a/(a+b)), lnb = log(b/(a+b));
        t = exp(a*lna)/a;
        u = exp(b*lnb)/b;
        w = t + u;
        if (p < t/w) x = pow(a*w*p, 1./a);
        else x = 1. - pow(b*w*(1.-p), 1./b);
    }
    afac = -gammln(a)-gammln(b)+gammln(a+b);
    for (j=0; j<10; j++) {
        if (x == 0. || x == 1.) return x;
        err = betai(a, b, x) - p;
        t = exp(a1*log(x)+b1*log(1.-x) + afac);
        u = err/t;
        x -= (t = u/(1.-0.5*MIN(1., u*(a1/x - b1/(1.-x)))));
        if (x <= 0.) x = 0.5*(x + t);
        if (x >= 1.) x = 0.5*(x + t + 1.);
        if (fabs(t) < EPS*x && j > 0) break;
    }
    return x;
}

};
const Doub Beta::EPS = numeric_limits<Doub>::epsilon();
const Doub Beta::FPMIN = numeric_limits<Doub>::min()/EPS;

```

CITED REFERENCES AND FURTHER READING:

- Abramowitz, M., and Stegun, I.A. 1964, *Handbook of Mathematical Functions* (Washington: National Bureau of Standards); reprinted 1968 (New York: Dover); online at <http://www.nrl.com/aands>, Chapters 6 and 26.[1]
- Pearson, E., and Johnson, N. 1968, *Tables of the Incomplete Beta Function* (Cambridge, UK: Cambridge University Press).